Lecture 21
t Intervals and Sample Size

Would you board this plane?

An airplane maker's study of its new passenger jet estimates the probability of a catastrophic in-flight failure. Your flight to the Bahamas is on one of these planes.

between 0.00012 and 0.0054 per flight

(1) Would you feel comfortable flying on this plane? Why or why not?
(2) If the maker could report this interval at 90%, 95% or 99% confidence, which would you want them to use? Why?
(3) Would you feel differently if the interval were narrower but had a lower confidence level?

0.00012 is 1 failure in 8,300 flights; 0.0054 is 1 in 185 flights
Airliners today: 1 fatal accident per 5.6 million flights (IATA, 2021–2025)
Even the best case is about 670 times riskier than a typical flight

Review

1 What does the confidence level C tell us?

The share of all intervals built this way that capture the parameter: 90%, 95%, 99%.

2 What is α?

The error rate, the share of intervals that miss: $\alpha = 1 - C$. At 95%, $\alpha = 0.05$, so 0.025 in each tail.

3 What is the critical value, and what does it do to the width?

$z_{1-\alpha/2}$, the number of SEs in estimate ± z × SE. Higher C → bigger z → wider interval.

Data: a gene and a drug

Clopidogrel (Plavix), a blood thinner used after heart attacks and stents, is switched on by the liver enzyme CYP2C19. People with the CYP2C19*2 variant switch on less of it.

X = 1 if a person carries at least one copy of *2, and 0 otherwise

PERSON	POPULATION	GENOTYPE	COPIES OF *2	X
HG00403	Southern Han Chinese	G/G	0	0
HG00404	Southern Han Chinese	A/A	2	1
HG00406	Southern Han Chinese	G/A	1	1
HG00407	Southern Han Chinese	G/G	0	0
HG00409	Southern Han Chinese	G/G	0	0
…	…	…	…	…

n = 504 East Asian participants of the 1000 Genomes Project, which sequenced 2,504 people worldwide to catalog human genetic variation

Example: carriers of CYP2C19*2

Estimate p, the proportion of all individuals of East Asian descent who carry the *2 variant.

277 of n = 504 people are carriers

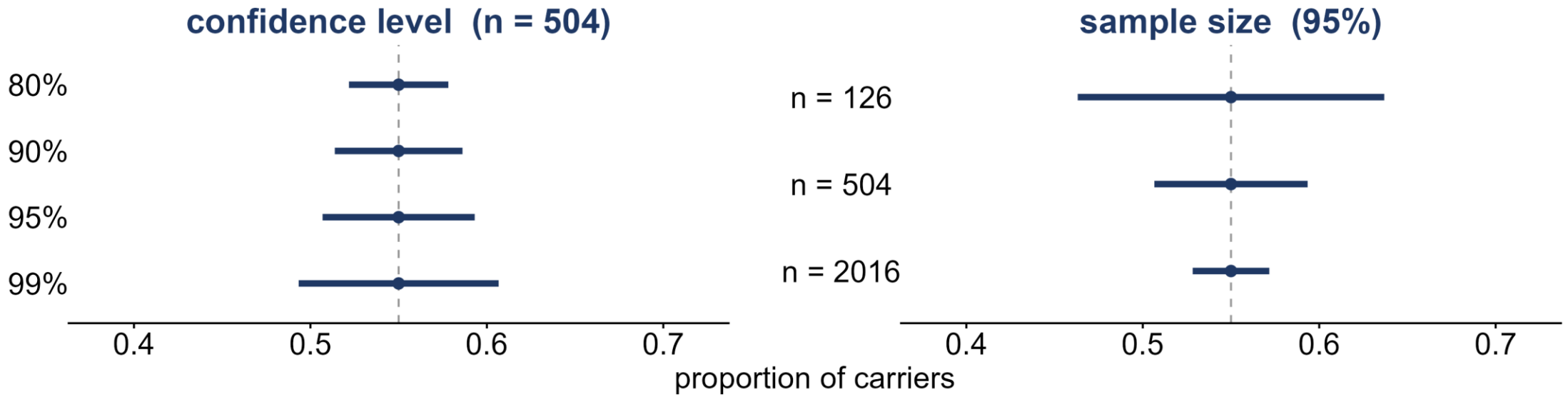
(a) What is $\hat{p}$? Is the check met? What is the standard error?
(b) What are the 90%, 95% and 99% confidence intervals for p?
(c) Can we say more than half of people carry *2?

(a) $\hat{p} = 277 / 504 \approx 0.550$; 277 and 227 are both > 5
$SE = \sqrt{(0.550 \times 0.450 / 504)} \approx 0.022$
(b) 90%: $0.550 \pm 1.645(0.022)$: 0.514 to 0.586
95%: $0.550 \pm 1.960(0.022)$: 0.507 to 0.593
99%: $0.550 \pm 2.576(0.022)$: 0.493 to 0.607
(c) At 90% and 95%, yes; the 99% interval includes 0.5

Making an interval narrower

The carrier interval ($\hat{p}$ = 0.550) at four confidence levels, and at 95% with 4 times fewer or more people.

What happens to the width as C goes up? As n goes up?



Higher C → wider. 4 times the n → half the width (SE has √n on the bottom)

What if σ is unknown?

Every interval for μ so far used σ, the population standard deviation.

$\bar{x} \pm z\, \sigma / \sqrt{n}$

(a) If we don't know μ, do we know σ?

(b) What could we use in its place?

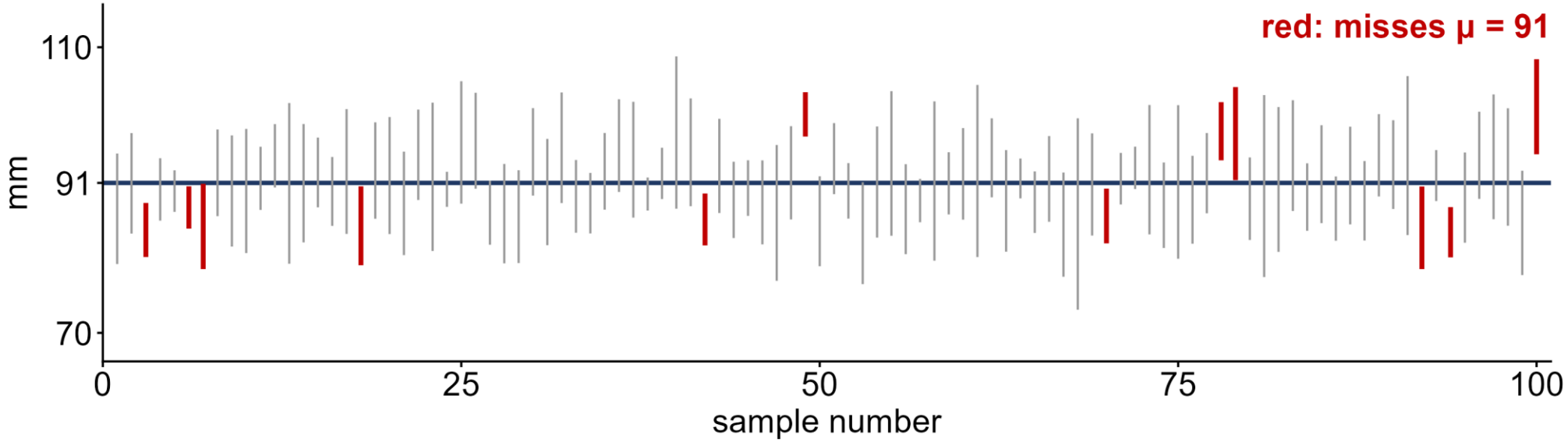
(a) Almost never

(b) s, the sample standard deviation: $\bar{x} \pm z\, s / \sqrt{n}$. Does that still work?

Does s in place of σ work?

Simulated: 100 samples of n = 5 whiskers from N(91, 8) mm; each line is $\bar{x} \pm 1.96\ s / \sqrt{5}$.

A 95% method should miss about 5 times. How many of these 100 miss μ?



12 of 100: s changes from sample to sample, so the intervals are too narrow too often

The t distribution

t distribution

a bell curve like N(0, 1) but with heavier tails; use it when s replaces σ

degrees of freedom

df = n − 1; the more df, the closer t is to N(0, 1)

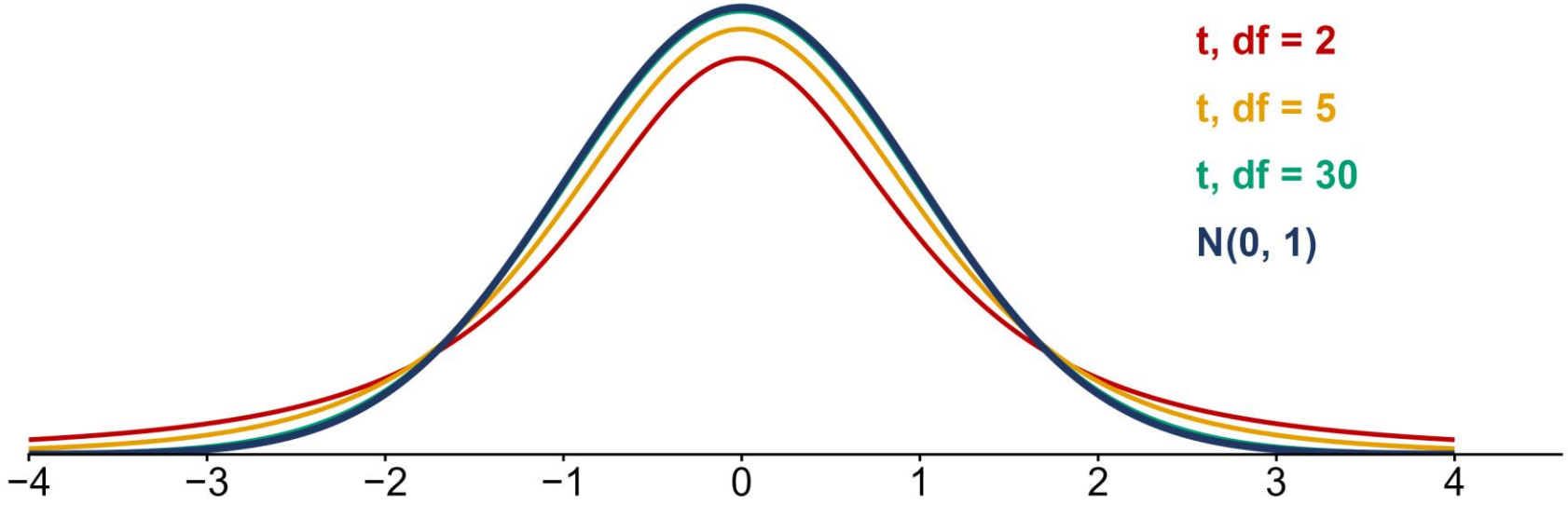
t critical value $t_{n-1,\,1-\alpha/2}$

from the t-table row for df = n − 1; larger than z, so the interval is wider

t versus the standard normal

Density curves of t with 2, 5 and 30 degrees of freedom, and the standard normal N(0, 1).

Which curve has the heaviest tails? What happens as df grows?



df = 2. As df grows, t approaches N(0, 1); by df = 30 they nearly match

Example: reading the t-table

In the course t-table, the row is df = n − 1 and the column is the confidence level.

95%, n = 10; 90%, n = 25; 99%, n = 10

(a) What is the t critical value for each?

(b) How does each compare with z for the same confidence level?

(a) df = 9: 2.262 df = 24: 1.711 df = 9: 3.250

(b) z = 1.960, 1.645, 2.576: t is always larger

Which interval?

a proportion p

$\hat{p} \pm z \sqrt{\hat{p}(1 - \hat{p}) / n}$, when $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$

a mean μ, σ known

$\bar{x} \pm z \sigma / \sqrt{n}$

a mean μ, σ unknown

$\bar{x} \pm t s / \sqrt{n}$, with $df = n - 1$

when the t interval works

the population is roughly normal, or $n \geq 30$

Data: seal whiskers

Ginter et al. (2012) measured the longest undamaged whisker of 92 seals and sea lions from 11 species, to study how whisker shape evolved.

X = the total length of a seal's longest whisker, in mm

SPECIES	SEALS (n)	MEAN LENGTH (mm)	SD (mm)
Harp seal	16	80.6	10.8
Harbor seal	15	86.5	10.5
Ringed seal	16	79.6	10.9
Spotted seal	9	91.2	7.9
Gray seal	8	73.8	12.5
…	…	…	…

One whisker per seal; only each species' n, mean and SD were published

Example: spotted seal whiskers

Estimate μ, the mean whisker length of all spotted seals, from the 9 that were measured.

n = 9; $\bar{x}$ = 91.2 mm; s = 7.9 mm

(a) z or t? How many degrees of freedom?
(b) What is the standard error?
(c) What is the 95% confidence interval for μ?

(a) t, because σ is unknown; df = 9 – 1 = 8, t = 2.306
(b) 7.9 / √9 ≈ 2.63 mm
(c) 91.2 ± 2.306(2.63) = 91.2 ± 6.06: 85.1 to 97.3 mm

Data: watermelon citrulline

Korea's national gene bank keeps seed of 100 watermelon varieties from around the world. Assefa et al. (2020) measured citrulline, the amino acid named after watermelon (Citrullus).

X = citrulline in a variety's fruit flesh, in mg per g of dry weight

VARIETY	ORIGIN	CITRULLINE (mg/g)
IT120008	Taiwan	27.16
IT199814	Kazakhstan	25.83
IT203017	Russia	26.38
IT203029	Kazakhstan	24.01
IT203627	unknown	32.79
…	…	…

A random sample of n = 12 of the 100 gene-bank varieties

Example: citrulline, 90% and 99%

Estimate μ, the mean citrulline of all 100 gene-bank varieties, from the sample of 12.

n = 12; $\bar{x}$ = 27.07 mg/g; s = 5.27 mg/g

(a) What are the degrees of freedom and the standard error?
(b) What are the 90% and 99% confidence intervals for μ?
(c) Which is wider, and why?

(a) df = 11; SE = 5.27 / √12 ≈ 1.52
(b) 90%: 27.07 ± 1.796(1.52): 24.34 to 29.80
99%: 27.07 ± 3.106(1.52): 22.35 to 31.79
(c) The 99% interval: more confidence needs a larger t

Choosing a sample size

margin of error m

the ± part of an interval: $m = z \times SE$, so a larger n gives a smaller m

n for a proportion

$n = z^2 p^*(1 - p^*) / m^2$, with $p^* = 0.5$ if there is no earlier guess

n for a mean

$n = (z \sigma / m)^2$, with σ from an earlier study

always round up

n = 59.1 means 60: rounding down misses the target m

Example: planning the next study

Before collecting data, decide how many to measure so the 95% interval is as narrow as you need.

carriers: m = 0.03, p* = 0.5; whiskers: m = 2 mm, σ ≈ 7.9 mm

(a) How many people for the proportion of *2 carriers?
(b) How many spotted seals for the mean whisker length?
(c) What happens to n if we want half the margin of error?

(a) $1.96^2\,(0.5)(0.5) / 0.03^2 = 1067.1$ → 1,068 people
(b) $(1.96 \times 7.9 / 2)^2 = 59.9$ → 60 seals
(c) 4 times as large: m is squared on the bottom

What you should know

critical value $z_{1-\alpha/2}$

1.645, 1.960, 2.576 for 90%, 95%, 99%; higher C → wider

σ unknown

use s and t with df = n – 1: $\bar{x} \pm t\, s / \sqrt{n}$

width

wider for higher confidence; narrower for larger n

sample size

$n = z^2\, p^*(1 - p^*) / m^2$ or $(z\, \sigma / m)^2$; always round up